

Design and Techno-Economic Analysis of a 15.30 kWp Off-Grid Solar Photovoltaic System for a Three-Storey Residential Building in Bilhari, Madhya Pradesh, India

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Abstract—This paper presents the complete engineering design and techno-economic evaluation of a 15.30 kWp off-grid solar photovoltaic (PV) system intended to supply a three-storey residential building located in Bilhari, Madhya Pradesh, India, with an average daily energy demand of 52.5 kWh. The design methodology covers electrical load assessment, PV array sizing, 48 V sealed-gel battery bank sizing, maximum power point tracking (MPPT) charge controller selection, inverter sizing, and development of a single-line diagram (SLD). The finalized configuration comprises 34 LG450 N2W-E6 modules (2 series $\times$ 17 parallel), five Victron SmartSolar MPPT 150/100 (48 V) charge controllers, a 48 V/1664 Ah HERE DEKA sealed-gel battery bank (4 series $\times$ 8 parallel, 32 units), and a 48 V DC/230 V AC pure sine wave inverter. A full-year hourly simulation carried out in PVsyst V8.1.3 predicts an annual useful energy delivery of 15,917 kWh, a performance ratio (PR) of 52.56%, a solar fraction of 83.16%, and a probability of loss of load (missing energy) of 7.0%. The techno-economic analysis, based on an initial capital investment of Rs. 30.09 lakh, indicates a simple payback period of 19.3 years, a levelized cost of energy (LCOE) of Rs. 11.55/kWh, and a 25-year return on investment (ROI) of 66.97%, with two battery replacements required at Years 10 and 20. The results confirm that the system is technically capable of meeting the residential demand with acceptable loss of load probability (PLOL = 5% design target), while highlighting that, under present lead-acid battery economics, off-grid systems of this scale are justified primarily by energy independence and reliability rather than short-term financial return.

Index Terms—Off-grid solar PV system, PV array sizing, battery bank sizing, MPPT charge controller, PVsyst simulation, techno-economic analysis, levelized cost of energy, rural electrification.

I. INTRODUCTION

Reliable electricity supply remains a challenge in several semi-urban and rural regions of India, where grid outages and voltage fluctuations are common. Off-grid solar photovoltaic (PV) systems with battery storage provide an autonomous alternative that decouples a residential load from the utility grid entirely, offering continuous power availability independent of grid conditions. This paper documents the complete design of a 15.30 kWp off-grid solar PV system developed for a three-storey residential building in Bilhari, Madhya Pradesh, India (23.1261°N, 79.9661°E, altitude 396 m), with an average daily consumption of 52.5 kWh.

The objective of this work is to develop and document a complete engineering design of the system, covering electrical load assessment, PV array sizing, battery bank sizing, MPPT charge controller selection, inverter sizing, single-line diagram (SLD) development, and techno-economic evaluation, so that the methodology can serve as a reproducible reference for similar residential off-grid installations [1].

II. LOAD ASSESSMENT

The daily electrical load of the residence was estimated by enumerating connected appliances across all floors, their rated power, average operating hours, and a diversity factor of 1.00 (i.e., no load coincidence discount was applied, representing a conservative design). Table 1 summarizes the load schedule.

Table 1. Residential Load Schedule

Load	Qty	Power (W)	Hrs/day	Energy (Wh/day)
LED Lights (G+1+2)	12	50	7.0	4200
Water Pump	2	1000	1.5	3000
Refrigerator	2	200	24	9600
Washing Machine	1	1350	2.0	2700
Domestic Appliances	1	1200	1.0	1200
Air Conditioner	2	1500	6.0	18000
Ceiling Fans	10	75	12	9000
Standby Loads	1	200	24	4800
Total		9500 W		52,500

The total connected load is 9.5 kW with a total daily energy demand of 52.5 kWh/day. The hourly load distribution (obtained from the PVsyst user-needs model) shows a pronounced early-morning peak from continuously running loads such as the refrigerator and standby consumers, and a secondary peak in the evening from lighting and air-conditioning use, which directly informs the required autonomy and battery depth-of-discharge assumptions used in Section IV.

III. PV ARRAY SIZING

The PV array was sized around LG Electronics LG450 N2W-E6 modules (450 Wp, monocrystalline). For the 52.5 kWh/day demand, the number of modules required is

$$N_{\text{modules}} = \frac{15,300}{450} = 34 \text{ modules.} \quad (1)$$

The finalized array configuration, consistent with the system single-line diagram and adopted throughout the design, uses **34 modules arranged as 2 modules in series $\times$ 17 parallel strings**, feeding five independent MPPT inputs (as detailed in Section V). This yields:

- Total PV capacity: 15.30 kWp
- Array V_{mpp} : ≈ 73.4 V
- Array V_{oc} (at -10°C): ≈ 108.6 V
- Total array current (I_{mpp}): ≈ 186.3 A

An alternative string configuration (3 series $\times$ 12 parallel, 36 modules, 16.20 kWp) was also evaluated during the design stage to assess higher string voltages against the 150 V MPPT input limit; this configuration yields a marginally higher installed capacity but was not carried forward, since the 2S $\times$ 17P/34-module arrangement matches the 48 V system bus more directly and was used for the final PVsyst simulation, cost estimate, and single-line diagram.

IV. BATTERY BANK SIZING

The battery bank was sized for the 48 V DC system bus using HERE DEKA Solar PV 8G8D sealed-gel lead-acid batteries (12 V, 208 Ah), with one day of autonomy, 80% depth of discharge (DoD), and a temperature correction factor of 0.90. The required battery energy and capacity are computed as:

$$E_{\text{req}} = 52.5 \times 1 = 52.5 \text{ kWh} \quad (2)$$

$$E_{\text{corr}} = \frac{52.5}{0.80 \times 0.90} = 72.92 \text{ kWh} \quad (3)$$

$$C_{\text{req}} = \frac{72.92 \times 1000}{48} \approx 1,520 \text{ Ah} \quad (4)$$

With 4 batteries required in series to reach 48 V and parallel strings of $1,520/208 \approx 8$, the total battery count is $4 \times 8 = 32$ units, giving an installed capacity of 1,664 Ah and a nominal stored energy of 79.87 kWh, of which 63.90 kWh (80% DoD) is usable. Table 2 summarizes the final battery bank design.

V. MPPT CHARGE CONTROLLER SIZING

Charge controllers were selected as Victron SmartSolar MPPT 150/100 units (150 V maximum PV input, 100 A maximum charging current), rated for a nominal controller power of $48 \text{ V} \times 100 \text{ A} = 4,800 \text{ W}$, with practical PV oversizing permitted to approximately 5.8 kWp per unit. The theoretical minimum number of controllers is

$$N_{\text{ctrl}} = \frac{15.30}{5.8} \approx 2.64 \Rightarrow 3 \text{ (minimum).} \quad (5)$$

For improved redundancy, reduced per-string current stress, and a one-to-one mapping with the five PV sub-arrays, the final design uses **five** Victron SmartSolar MPPT 150/100 controllers, each handling 3.06 kWp and an operating current of 36.8 A against a 100 A rated capacity—a safety margin of 63.2 A per controller.

Table 2. Battery Bank Design Summary

Parameter	Value
Battery model	HERE DEKA Solar PV 8G8D
Rating	12 V, 208 Ah (Sealed Gel)
Configuration	4 Series $\times$ 8 Parallel
Total batteries	32
System voltage	48 V DC
Installed capacity	1,664 Ah
Nominal energy	79.87 kWh
Usable energy (80% DoD)	63.90 kWh
Days of autonomy	1

Table 3. PVsyst Annual Simulation Results

Parameter	Value
Available solar energy	25,375 kWh/yr
Useful energy from solar (E_User)	15,917 kWh/yr
Excess energy (unused, battery full)	8,412 kWh/yr
Missing energy (loss of load)	3,245 kWh/yr
Energy need of the user (load)	19,163 kWh/yr
Specific production	1,040 kWh/kWp/yr
Performance ratio (PR)	52.56%
Solar fraction (SF)	83.16%
Loss-of-load time fraction	7.0%
Battery ageing —cycling SOW	54.7%
Battery ageing —static SOW	92.9%

VI. SYSTEM ARCHITECTURE AND SINGLE-LINE DIAGRAM

Fig. 1 shows the complete single-line diagram (SLD) of the system. The 34-module PV array is divided into five sub-arrays (2S $\times$ 17 strings distributed across the five MPPT inputs), each protected by a 100 A DC fuse before reaching the DC combiner and busbar. Two 200 A DC disconnectors isolate the combiner bus from the 48 V DC/230 V AC pure sine wave inverter. The inverter output feeds the AC distribution board through an AC MCB/isolator, supplying lighting, fan, socket, and appliance circuits. Earthing is provided at the inverter, distribution board, and all exposed metallic parts in accordance with standard electrical safety practice.

VII. PERFORMANCE SIMULATION USING PVSYST

The finalized 34-module (15.30 kWp) array, together with a simplified battery-pack model in PVsyst (Solar PV 8G8D, 6 parallel $\times$ 4 series, 24 units, 1248 Ah), was simulated over one year at an hourly time step using Meteonorm 9.0 synthetic weather data for the project site, with a fixed tilt/azimuth of $25.5^\circ/0^\circ$ and an albedo of 0.20. Table 3 presents the key simulation outputs.

The PVsyst loss diagram indicates that the largest single loss term is unused (excess) energy when the battery bank is full, at 31.98% of the array's virtual energy at MPP, followed by PV thermal losses (-9.77%) and converter operating losses (-3.68%). Battery round-trip efficiency loss accounts for a further 6.06% of stored energy. Table 4 shows the monthly energy balance, highlighting that the loss of load is concentrated in

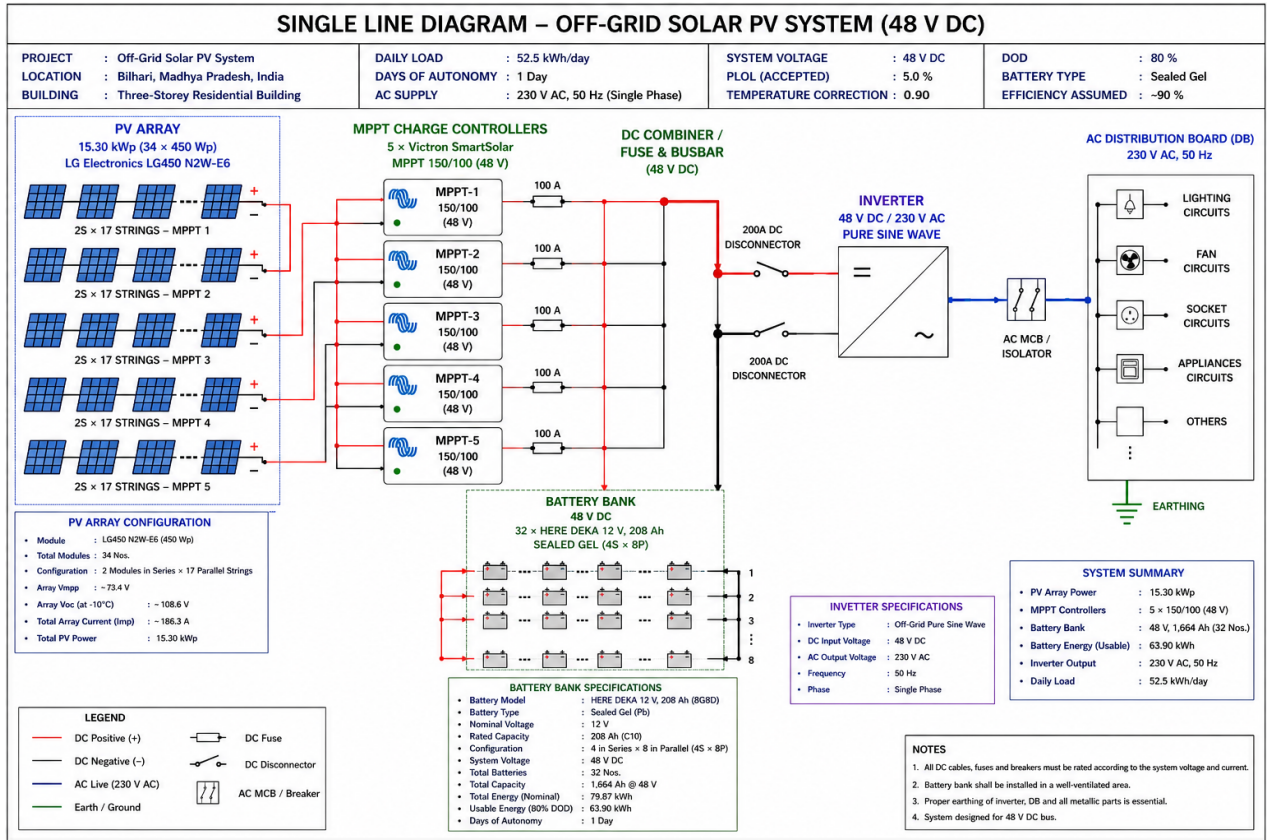


Figure 1. Single-line diagram of the 15.30 kWp off-grid solar PV system (48 V DC bus).

Table 4. Monthly Energy Balance (PVsyst)

Month	GlobEff (kWh/m ²)	E_Avail (kWh)	E_Miss (kWh)	SolFrac
Jan	175.5	2371	224	0.874
Feb	173.0	2289	179	0.879
Mar	204.7	2653	162	0.900
Apr	196.8	2507	151	0.903
May	197.5	2490	154	0.900
Jun	143.4	1860	225	0.864
Jul	98.6	1285	604	0.629
Aug	95.9	1260	592	0.622
Sep	139.8	1837	309	0.817
Oct	177.5	2315	245	0.850
Nov	164.4	2189	186	0.881
Dec	172.1	2317	213	0.869
Year	1939.1	25375	3245	0.832

Table 5. Initial Capital Cost Breakdown

Component	Qty	Cost (Rs.)
LG 450 W PV modules	34	425,000
HERE DEKA batteries	32	1,344,000
Victron MPPT 150/100	5	310,000
48 V, 15 kVA inverter	1	300,000
Mounting structure	1 lot	180,000
DC combiner box	1	40,000
AC distribution board	1	30,000
Cables & connectors	1 lot	120,000
Lightning protection	1	35,000
Earthing system	1	25,000
Installation & commissioning	1 lot	200,000
Total		3,009,000

0.77, the estimated annual gross PV generation is

$$E_{\text{annual}} = 15.3 \times 5.5 \times 365 \times 0.77 \approx 23,640 \text{ kWh/yr}, \quad (6)$$

the July–August monsoon period, when global effective irradiation drops to 95–99 kWh/m², driving the monthly solar fraction down to 0.62–0.63 compared with 0.86–0.90 in the drier months.

VIII. TECHNO-ECONOMIC ANALYSIS

The initial capital cost of the system, summarized in Table 5, totals Rs. 30.09 lakh (Rs. 3,009,000), of which the 32-unit battery bank alone represents nearly 45% of the total investment. Assuming 5.5 peak sun hours/day and a performance ratio of

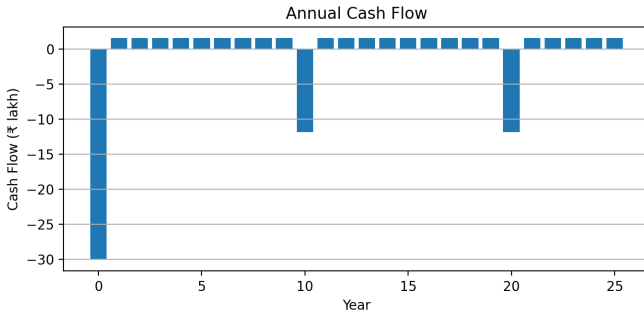
which, at a retail electricity tariff of Rs. 8.5/kWh, corresponds to gross annual savings of Rs. 2.01 lakh. After deducting operation and maintenance costs (1.5% of capital cost, Rs. 45,135/yr), the net annual saving is Rs. 1.56 lakh/yr, giving a simple payback period of

$$T_{\text{payback}} = \frac{30.09}{1.558} \approx 19.3 \text{ years}. \quad (7)$$

It is noted that this simplified annual-generation estimate (23,640 kWh/yr, based on standard PSH/PR assumptions) is

Table 6. Techno-Economic Summary

Parameter	Value
Total investment	Rs. 30.09 lakh
Annual generation (estimated)	23,640 kWh
Annual gross savings	Rs. 2.01 lakh
Annual O&M cost	Rs. 45,135
Net annual savings	Rs. 1.56 lakh
Simple payback period	19.3 years
Payback (with hypothetical subsidy)	18.8 years
Battery replacement	Years 10 and 20
Levelized cost of energy (LCOE)	Rs. 11.55/kWh
Lifetime energy delivered	591 MWh
25-year return on investment	66.97%

**Figure 2.** Projected annual cash flow over the 25-year project life.

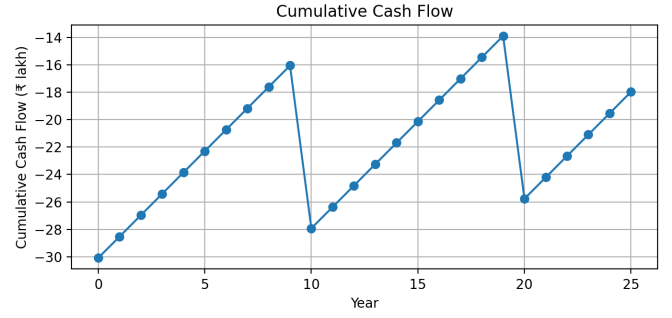
somewhat higher than the PVsyst-simulated energy actually delivered to the load (15,917 kWh/yr), the difference being attributable to battery-full curtailment, monsoon-season shortfalls, and conversion losses captured in the hourly simulation but not in the simplified estimate; the economic figures should therefore be read as an upper-bound, first-pass estimate.

Since the battery bank has an expected service life of 10 years, two full replacements (Rs. 13.44 lakh each) are required at Years 10 and 20. Over the 25-year project life, the life-cycle cost is Rs. 30.09 lakh (capital) + Rs. 26.88 lakh (replacements) + Rs. 11.28 lakh (O&M) = Rs. 68.25 lakh, against a lifetime energy delivery of 591 MWh, giving a levelized cost of energy of

$$\text{LCOE} = \frac{68.25 \text{ lakh}}{591,000 \text{ kWh}} \approx \text{Rs. } 11.55/\text{kWh}. \quad (8)$$

The 25-year lifetime savings of Rs. 50.24 lakh give a return on investment of 66.97%. Table 6 summarizes the key economic indicators. Note that the PM Surya Ghar rooftop solar subsidy does not apply to this off-grid configuration; a hypothetical subsidy of Rs. 78,000 would reduce the payback period only marginally, to 18.8 years.

Fig. 2 and Fig. 3 present the projected annual and cumulative cash flows over the 25-year project life. The cumulative cash flow curve remains negative throughout the project life under the assumed Rs. 8.5/kWh tariff, reflecting the fact that, for lead-acid battery-based off-grid systems with two full battery replacements, energy independence and reliability—rather than short-term financial return—are the principal justification.

**Figure 3.** Projected cumulative cash flow over the 25-year project life.

IX. RESULTS AND DISCUSSION

The completed design meets the residential energy requirement of 52.5 kWh/day with a single day of autonomy and an accepted loss-of-load probability design target of 5%; the full hourly PVsyst simulation shows a realized loss-of-load time fraction of 7.0%, concentrated almost entirely in the July–August monsoon months, when solar fraction falls to approximately 0.62–0.63. Outside the monsoon period, the solar fraction consistently exceeds 0.85, and in several months exceeds 0.90, indicating that the system is well matched to the load for the majority of the year. The relatively low overall performance ratio (52.56%) is dominated by unused/curtailed energy (32% of array output at MPP) once the battery bank reaches full state of charge, which is a structural characteristic of a fixed-capacity, non-grid-connected battery system rather than a design deficiency; it indicates headroom for either a reduction in installed PV capacity or the addition of dump/deferrable loads (e.g., day-time water heating) to make productive use of the otherwise-curtailed energy.

From an economic standpoint, the 19.3-year simple payback period and 66.97% 25-year ROI are consistent with published experience for lead-acid battery-based off-grid residential systems in India, where the battery bank represents close to half of capital expenditure and requires replacement roughly every decade. Migration to lithium iron phosphate (LiFePO₄) batteries, which offer longer cycle life and higher usable DoD, would be expected to materially improve both the LCOE and payback period, at the cost of higher upfront battery expenditure.

X. CONCLUSION

This paper has presented the complete design and techno-economic evaluation of a 15.30 kWp off-grid solar PV system for a three-storey residential building in Bilhari, Madhya Pradesh, India. The system, comprising a 34-module (2S×17P) PV array, five Victron SmartSolar MPPT 150/100 charge controllers, a 32-unit (4S×8P) 48 V/1664 Ah HERE DEKA sealed-gel battery bank, and a 48 V/230 V pure sine wave inverter, was verified through hourly PVsyst simulation to deliver 15,917 kWh/yr to the load at a performance ratio of 52.56% and a solar fraction of 83.16%, with a loss-of-load time fraction of 7.0%, close to the 5% design target. The techno-economic analysis indicates an initial investment of Rs. 30.09 lakh, a simple payback of 19.3 years, and an LCOE of Rs. 11.55/kWh, confirming that, while the system is technically robust, its adoption in the present lead-acid battery paradigm is justified chiefly

by energy security and reliability. Future work should examine lithium-based storage chemistries, load-shifting/dump-load strategies to reduce curtailment losses, and hybrid grid-interactive configurations to improve overall economic performance.

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